

Financial Decision Buddy: A Decision-support Tool to Bridge the Gaps in Financial Education

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ABSTRACT

SDG 4 education and SDG 1 poverty are issues of global concern. Young people may fall into poverty due to various reasons: student debt, unemployment, and unpredictable events such as the pandemic. Therefore, education to improve their financial competence is necessary to increase their resilience to those risks and pressures. We analysed 62 participants' questionnaire data and found that most participants struggle with making optimal financial decisions when spending. With this in mind, we designed a system that uses machine learning to analyse users' previous spending habits and models it against a decision tree framework to calculate the most optimal decision within users' budgets. The system used affective interaction and anthropomorphic communication to educate users in real-time shopping environments, for example when they're shopping in a grocery store. Our user testing demonstrated that the design was helpful with cognitive offload and encouraged critical thinking when making purchasing decisions. Furthermore, our design idea sheds light on possible approaches to applying virtual AI assistants to improve skills when lacking resources such as time, place, or human experts.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Interaction design.**

KEYWORDS

financial education, decision support, affective interaction, MEU framework, financial literacy, reducing poverty

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1 INTRODUCTION

Education to raise young people's financial competency has become more and more crucial these days, which ties to the UN's sustainable goals Number 1 (No Poverty) and Number 4 (Quality Education) [12]. Due to the recent pandemic, the global economy slumped [2] and the youth unemployment rate raised to 14.9% in 2020 [14], making it difficult for young people to get jobs and earn a living.

Besides such unexpected events, young people also face financial pressure just from education alone. For example, the average student loan in England has risen to 45,000 pounds, increasing by 75% compared to the pre-2012 system [3] putting graduates into debt even before getting a job. If young adults don't manage their finances well, they could spiral into further debt and poverty. However, only 38% of children and young people in the UK have been taught financial education at school [18], suggesting that some young people may be disadvantaged when faced with various financial risks and pressure.

According to our conducted survey, most young people struggled to manage their daily expenses, specifically in the areas of entertainment and food such as personal and grocery shopping, while some express having high financial pressure. These issues could cause young people to make sub-optimal financial decisions, thus putting them at risk of spending over their means and in the long run, poverty. Although young people are aware of this, they often find it too complicated or difficult to improve their financial skills, specifically in the areas of budgeting and savings.

Based on the financial competence framework [19], this is tied to two major parts of individuals' financial literacy: Knowledge and Money & Financial Management. These two aspects are linked to people's ability to make well-thought-out financial decisions in consideration of the overall utility such as their budget and the product's quality.

To prevent young people from falling into poverty, and to raise their overall financial well-being, we designed a product to bridge the education gaps in financial literacy by providing support to real-time purchasing decisions and short-term budgeting. Furthermore, using innovative interactions and the application of AI assistance, we expect our product to help develop young people's financial literacy in the long run.

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2 RESEARCH

2.1 Literature Research

2.1.1 Decision making and its possible improvement. It is undeniable that decision-making is a relatively complicated process. People may misjudge or be misdirected due to limited cognitive capacity, thus leading to heuristics and biases [17]. This limitation of human judgment and decision-making is used widely. For example, Crisp [6] pointed out that persuasive advertising was misusing persuasive techniques, which would result in consumers being misdirected from making decisions based on their merit. Therefore, we adopted the Maximum Expected Utility (MEU) Framework that can calculate products' overall value and utility [13], which can provide most optimal choices out of all possible alternatives.

2.1.2 Implementing AI on teaching and real-time support. The idea of using AI to assist or even replace human experts has been developed since the 1960s. Thomson and Willoughby [16] developed a web-based Expert System (ES) to give evidence-based advice on herbicide use. Users could input their needs and conditions, and the system would give suggestions based on options' utilities. Besides consulting, ES was also tested on its capacity to teach. Chan et al. [5] designed an ES with the knowledge of family planning. It simulated medical cases and required students to give their diagnoses to compare with ES's answer. This ES showed significant advantages in explaining and stimulating learning. Students also reported that learning with ES was interesting and helpful.

2.1.3 Affective interaction and anthropomorphism. Although AI can be a powerful tool, there are still obstacles compared to a human consultant. For example, it was found in automated driving that people trust AI's decision less than trust a human's decision [10]. Affective interaction can be a possible solution to this. In human-robot interactions, emotion motivates intuitive dialogue and better communication [15]. Buchholz et al. [4] found that human teammates perceived it as trustworthy and willing to cooperate with it when computers used self-blaming strategies. In addition to emotions, automatic agents with anthropomorphic design would elicit more trust with the same advice quality compared to computers without anthropomorphic design [11].

2.2 User Research

2.2.1 Identifying users' struggles. We started by identifying young people's struggles with financial management. A user survey was conducted with 62 young adults between 18 and 29 across Europe and Asia. This survey focused on young people's financial situation, spending behaviour, and behaviour related to financial management.

According to the result, although most participants believed they were spending rationally, 37% agreed that they struggled to manage their finances. In addition, 51% of young adults felt their ability to manage their purchasing was inadequate, specifying the lack of self-control or planning. In which the majority mentioned that they struggled with daily expenses like food spending, and less than 50% of the participants affirmed that they only bought what they needed in grocery stores. On top of that, almost 50% participants mentioned needing help with budgeting and saving when they started managing their finances.

An affinity map was created to analyse the answers we collected from open-ended questions (figure 1). When asked what they found difficult, besides balancing their requirements and necessities, participants mentioned issues such as difficulty in tracking expenses, lack of good strategy for budgeting and saving, and impulsive buying due to persuasive advertisement. The reason behind these struggles can be summed up into three aspects: 1) lack of efficient tracking strategy: Most of our participants frequently forget to record and track their spending from time to time; 2) hard to balance between needs and wants: participants were struggled on to what extent they could satisfy their desires outside necessities became a problem; 3) unable to evaluate options in their utility: participants often made judgments without well-justified reasons, for instance, several participants mentioned irrational purchasing due to discounts or thought they might need it one day while ignoring the real situation.

The user research provided insights into our design direction. As a result, we decided to create a product that could help young people keep track of their in-store purchases while providing suggestions based on the options' utilities and the users' preferences.

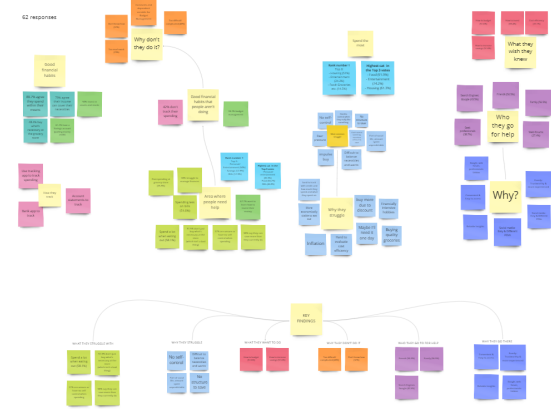


Figure 1: The affinity map created based on questionnaire answers.

2.2.2 Establishing user requirements. Before ideating, we conducted follow-up user interviews with five participants on their product expectations. We collected participants' thoughts on where they would find help from and what aspects they care about regarding the support they got. Overall, we established six user-reported requirements:

- Informative and functional: Suggestions need to be explainable and well-justified.
- Trustworthy and Personalised: Whatever our product would be, it needs to elicit trust from users and know them well.
- Easy to use: The support and suggestions need to be easy to understand and straightforward.
- Entertaining: The interaction need to be engaging and fun.
- Aesthetically pleasing: The interfaces need to be pretty and clean.
- Innovative: The interaction need to be creative.

3 DESIGN PROCESS

3.1 Existing Designs

In our user survey, we noticed that 70% of the participants who consciously kept track of their spending relied on mobile finance apps. Based on this finding, we evaluated several widely used finance apps. The app that closely resembled the concept we were going for was Fortune City.

It is one of the most widely used trackers in the App Store. This app has three main features: 1) Spending categories, 2) Monthly reports based on records, and 3) Allowing users to set a plan for future spending. One main issue with this app was the need for manual input, which might result in omission or inaccuracy. Fortune City also gamified the action of recording spending by allowing users to build cities. Although this gamification is used to motivate self-tracking, we can also adopt this concept to motivate budgeting and good decision-making.

3.2 Initial Ideas

With the initial user requirements in mind, we created a persona to help visualise our target user and devised a criteria matrix containing an extended list of 10 criteria. Each criterion was weighted according to the users' needs and the aim of our project. The matrix attached the greatest importance to being informative and functional, serving our primary goal of aided learning, specifically in financial decision-making. Other criteria such as Ease of use, Innovation, Entertainment, User Stickiness, and Aesthetics, were also ranked highly in the matrix as these may raise the users' motivation and experience substantially. We then conducted a Crazy 8s activity to generate different ideas. As a team, we selected our favourite ideas and evaluated them against the criteria matrix, the following three preliminary ideas were selected as the best designs for further development.

- **An Affective Robot Dog** that can provide real-time feedback through emotional reactions (physical interactions, facial expressions, etc.) The interface on its body also helps users gauge their spending and check if their decisions are optimal.
- **Simulation Game** that would gamify the processes and outcomes of decision-making. When users save money in the real world, they will get an equal amount of in-game currency that can be used to build up and decorate their virtual house and garden.
- **SPENDSMART** An AI learning system that analyses users' spending habits through their purchasing history. Users feed their bills and receipts into the system, along with their budgets and preferences. The AI will then provide suggestions to help users make better decisions according to their goals. The rapid development of digital technologies and machine learning techniques have made it possible to analyse consumer behaviours and predict demands at the individual level [1] [8], which is the foundation of applying AI analytics in our design.

The Affective Robot Dog was ranked the highest in our evaluation. Its real-time decision support feature and use of affective interaction could help elicit trustworthiness. However, it was lacking

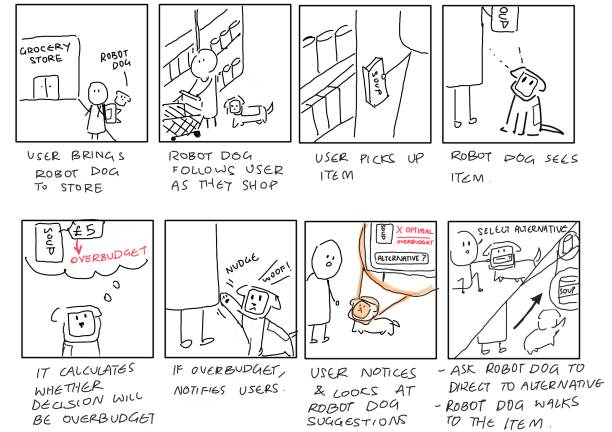


Figure 2: The storyboard of how the robot dog would function as a decision supporter.

in areas such as educational and long-term entertainment, therefore, we decided to add on features from the Simulation Game and SPENDSMART concept. The gamification could make our design more entertaining, while the AI could provide personalised and explainable suggestions through an expert system.

Combining what we got, we settled down on the core concept of our initial design. It included a physical robot dog with an interactive face. The robot dog follows the user as they shop in-store and tracks their behaviour through a built-in camera. When there was a real-time suggestion, the dog would use physical interactions to alert users to check its face for suggestions.

3.2.1 User feedback on initial design. We tested the concept of the affective dog (figure 2) with 5 participants. According to participants' feedback, the affective design was considered favourable and pleasant. However, some feedback showed that our design could be too excessive. The majority enjoyed the idea of an "animal consultant", but some participants worried that interacting with a robot dog in public could harm their privacy. Additionally, some participants also mentioned that having a physical robot dog that follows them around could be "annoying". Several participants also questioned whether this design was affordable.

Based on this feedback, we decided to build our design on a more portable and affordable platform, while keeping the physical "animal companion" interaction as much as possible. Due to this, we decided to further develop the robot dog into a hybrid system.

4 FINAL DESIGN

Our final design consists of three parts (figure 3).

- An app which is supported by an expert system and machine learning which can provide real-time suggestions without being too prominent.
- A physical extension that includes a phone case with movable ears that could be clipped onto the phone, fulfilling the physical and affective interaction.

- A pendant that would identify users' selections and provide haptic feedback in the form of vibrations. This haptic feedback would also nudge users to check the app for suggestions on how to improve their decisions.

The app consists of behaviour analysis, real-time decision support, feedback reports, and community forum and reviews. Users can customise their "Financial Decision Buddy" in the app based on their preferences. The "Decision Buddy" is powered by the AI algorithm (ES system and machine learning), which can actively take users' personalised factors into consideration during real-time shopping. Through this approach, our "Decision Buddy" can give recommendations and suggestions that meet users' expectations more flexibly and accurately, which precedes just a recommended system. Like the robot dog, we used animal figures as the avatars to represent our Expert System. The "talking animals" were familiar anthropomorphic figures [7]. By doing this, we expected our design to be more trustworthy and likeable.

When under shopping mode, users will first key in the shop they're at onto the app. This will allow the AI to pull up the shops' database to help with the item cost tracking, users will then slot their phone into the ear extension phone case to mimic the physical appearance of their avatar (in the example, it would be a dog named Nala). As users put items into their shopping cart, the pendant would visually identify the item, automatically track the cost and analyse the decisions based on ES's judgment.

This real-time information will be visually shown on their avatars' faces to keep users informed. Using the on-screen facial expressions and movable ears, the avatar will express emotions based on how optimal the users' decisions are. When a sub-optimal decision puts the user at risk of over-budgeting, users will be provided haptic feedback on the pendant, and the avatar's ears and emotions will drop. Users will also be provided suggestions on how to re-optimize their cart such as alternative product options that are more suited to their budget.

After each shopping trip, the app will reward users based on how optimal the users' decision-making was. This reward is for the in-app game, where users can interact with their budget buddy outside the shopping companion function. The app will also provide a report that includes an evaluation of users' recent spending behaviour and educates users on what they did well and what could be improved.

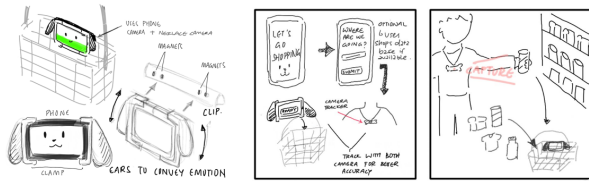


Figure 3: The final design idea was based on the iteration of the robot dog.

4.1 Usability Testing

To test our design, we conducted user tests with 5 participants. Three of them were recruited from the previous user interviews,

and two were new participants who got to know our Financial Decision Buddy for the first time.

Our prototype included a phone to display the interface, a phone case with manually adjustable cardboard ears, an Apple Watch to simulate the haptic feedback necklace and a "decision-tracking" software. The software was written in Python and simulated the functions of the financial decision buddy, such as adding and removing items from the cart, cost tracking, and changes in facial expressions. An "Ask" button was also implemented on the software's interface to mimic the real-time decision support function.

Our user test was separated into 3 tasks:

Task 1: Participants completed a shopping task for a persona in a mocked supermarket scenario. They were given the persona's budget, needs, and preferences, and show a variety of items ranging in cost and quality. Participants had to identify the items that would not only fit the budget but also make the persona as happy as possible. This was completed without the prototype to see how users would shop in a regular setting with a set budget and with personal preference.

Task 2: Participants did a walk-through of the app using a mixed-fidelity interactive prototype. This helped participants better understand how the app works and to see if the user flow was intuitive.

Task 3: Participants re-did the shopping task in section 1, but for a different persona and with the Financial Decision Buddy's help. We simulated this using a Wizard of Oz method (figure 4). One team member observed participants, while the other two members controlled the software and Apple Watch to simulate the behaviour-tracking function and manually moved the ears on the phone case to simulate the ear extension. When the avatar's emotion dropped, participants could use the "Ask" button for suggestions provided by another team member. At the end of the mock shopping, participants were provided with a verbal report of their performance, including brief feedback and an analysis of their decisions based on the decision analysis strategy.

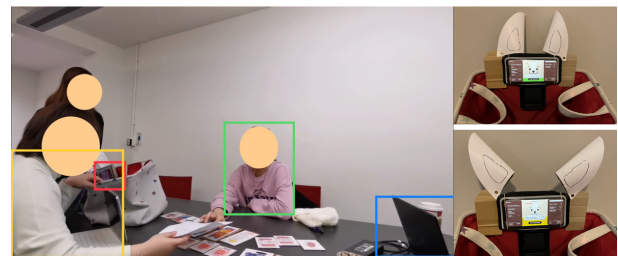


Figure 4: The user test setting. The green square (in the middle) was a participant wearing a vibration pendant. The ear extension of the prototype will be manually adjusted based on Nala's emotions. From left to right, the yellow square (left 1) was the team member simulating real-time communication and suggestion function. The small red square (left 2) was the prototype that was presented on the phone. The blue square (right 1) was the team member operating the prototype on the phone.

4.2 User Feedback

After the experiment, we collected feedback from participants via interviews. Our prototype rated 85% against the criteria matrix. When asked about participants' experiences in task 1, participants diverged in perceived difficulty. However, all participants agreed that our design helped boost their confidence in decision-making and made the process more entertaining. Additionally, the ear extension attracted participants' attention effectively, while the avatar's emotions pushed them to think more critically about their decision. Regarding our goal of skill development, participants affirmed that our design would motivate them over time to build better spending habits.

We collected users' feedback on the interface design in the follow-up interview. Participants stated that the ear extension and avatar's emotions were visually appealing and easy to understand. The colour variations that reflected the avatar's emotional changes were also engaging and captivating. In terms of information clarity, most participants rate the interface as very clear. However, two participants mentioned that the text font might need to be bigger so it can be easier to see for some users.

Nevertheless, there was also feedback that showed room for improvement. Two participants preferred real-money incentives over gamification rewards, while another participant would use the financial decision buddy as a decision machine that can decide for them instead of a decision expert that can help them with their own decision.

5 CONCLUSION AND FUTURE WORKS

We designed a product to provide users with real-time support and follow-up feedback on purchasing decisions. This design uses AI and affective interaction design to introduce a more trustworthy and accessible "consultant", which can help participants improve their financial decision-making skills more entertainingly and effortlessly, and could hopefully improve their skills while reflecting on information and judgment critically. In the long run, this could lead to better financial competency, and in turn, reduce the risk of poverty. Moreover, affective AI assistance in skill learning can be a promising solution in situations with limited human resources.

There were also limitations in our design. While our design is trying to combine AI-assisted learning with practice in real-world scenarios, users could become over-reliance on it. Grinschgl et al. [9] found that offloading might lead to worse cognitive capacity. However, the same study also found that raising awareness of upcoming cognitive tests almost counteracted the negative effect of offloading. This finding indicates that the new design can set different levels of "challenge" depending on how much assistance users require from the system. The app would provide better incentives such as cash or shopping vouchers to motivate users to 'level up' to higher self-dependency.

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